

# A First Course in Causal Inference Ch 6

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## Ch. 6 Randomization and Regression Adjustment

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- ▶ Stratification and post-stratification is for design and analysis with discrete covariates.
- ▶ How should we deal with multidimensional, possibly continuous covariates?

|                    | design          | analysis              |
|--------------------|-----------------|-----------------------|
| discrete covariate | stratification  | analysis              |
| general covariate  | rerandomization | regression adjustment |

## 6.1 Rerandomization

- ▶ Setup: a finite population of  $n$  units;  $n_1$  units in a treatment group;  $n_0$  in a control group;  $\mathbf{Z} = (Z_i, \dots, Z_n)$  is the treatment vector for these units; and unit  $i$  has covariate  $X_i \in \mathbb{R}^{K^1}$ .
- ▶ Stack them as an  $n \times K$  covariate matrix  $\mathbf{X} = (X_1, \dots, X_n)^T$  and center them as mean zero  $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i = 0$ .

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<sup>1</sup>This covariate  $X_i$  can have continuous or binary components

## 6.1 Rerandomization

- ▶ In a CRE, the covariates in the treatment and control groups are balanced **on average** (*not for every stratum!*). Namely, let

$$\widehat{\tau}_X = \frac{1}{n_1} \sum_{i=1}^n Z_i X_i - \frac{1}{n_0} \sum_{i=1}^n (1 - Z_i) X_i,$$

then  $\mathbb{E}[\widehat{\tau}_X] = 0$ .

- ▶ In the realized treatment allocation, the realized value of  $\widehat{\tau}_X$  is often not zero.
- ▶ Building on Neyman (1923),

$$\text{cov}(\widehat{\tau}_X) = \frac{S_X^2}{n_1} + \frac{S_X^2}{n_0} = \frac{n}{n_1 n_0} S_X^2, \quad (1)$$

where  $S_X^2 = \frac{1}{n} \sum_{i=1}^n X_i X_i^T$  is the finite-population covariance matrix of the covariates.

## 6.1 Rerandomization

### Definition 6.0 Mahalanobis Distance

Given a probability distribution  $Q \in \mathbb{R}^N$ , with mean  $\boldsymbol{\mu} = (\mu_1, \dots, \mu_N)^T$  and positive semi-definite covariance matrix  $\boldsymbol{\Sigma}$ , the Mahalanobis distance of a point  $\mathbf{x} = (x_1, \dots, x_N)^T$  from  $Q$  is defined as:

$$M(\mathbf{x}, Q) = \sqrt{(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu})}.$$

Let  $\mathbf{y} = (y_1, \dots, y_N)^T$  be another point, then the Mahalanobis distance between  $\mathbf{x}$  and  $\mathbf{y}$  is defined as:

$$M(\mathbf{x}, \mathbf{y}) = \sqrt{(\mathbf{x} - \mathbf{y})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \mathbf{y})}.$$

## 6.1 Rerandomization

Following Def. 6, the difference between the treatment and control groups is

$$M = \hat{\tau}_X^T \text{cov}(\hat{\tau}_X)^{-1} \hat{\tau}_X = \hat{\tau}_X^T \left( \frac{n}{n_1 n_0} S_X^2 \right)^{-1} \hat{\tau}_X.$$

- ▶  $M$  is meaningful only if  $S_X^2$  is invertible.
- ▶  $M$  is invariant under non-degenerate linear transformation of  $X$ .

### Lemma 6.1

$M$  remains the same if we transform  $X_i$  to  $b_0 + BX_i$  for all units  $i = 1, \dots, n$  where  $b_0 \in \mathbb{R}^K$  and  $B \in \mathbb{R}^{K \times K}$  is invertible.

▶ Proof

## 6.1 Rerandomization

- ▶ With large  $n$ , the Mahalanobis distance  $M$  is approximately  $\chi_K^2$  under the CRE.
- ▶ Rerandomization avoids covariate imbalance by discarding the treatment allocations with large values of  $M$ .

### Definition 6.1 ReM(Rerandomization)

Draw  $\mathbf{Z}$  from CRE and accept it if and only if

$$M \leq a,$$

for some predetermined constant  $a > 0$ .

- ▶ When  $a = \infty$ , we just conduct a CRE.
- ▶ When  $a = 0$ , there are very few feasible treatment allocations, then consequently, the experiment has little randomness.
- ▶ In practice, we choose a small but not too extreme  $a$  or some upper quantile<sup>8/17</sup>

## 6.1 Rerandomization

- ▶ When  $a = \infty$ , we just conduct a CRE.
- ▶ When  $a = 0$ , there are very few feasible treatment allocations, then consequently, the experiment has little randomness.
- ▶ In practice, we choose a small but not too extreme  $a$  or some upper quantile of a  $\chi_K^2$  distribution.

## 6.1 Rerandomization

### Condition 6.1

As  $n \rightarrow \infty$ ,

- ▶  $\frac{n_1}{n}$  and  $\frac{n_0}{n}$  have positive limits;
- ▶ the finite population covariance of  $\{X_i, Y_i(1), Y_i(0), \tau_i\}$  has a finite limit;
- ▶  $\max_{1 \leq i \leq n} \frac{\{Y_i(1) - \bar{Y}(1)\}^2}{n} \rightarrow 0$ ,  $\max_{1 \leq i \leq n} \frac{\{Y_i(0) - \bar{Y}(0)\}^2}{n} \rightarrow 0$  and  $\max_{1 \leq i \leq n} \frac{X_i^T X_i}{n} \rightarrow 0$ .

## 6.1 Rerandomization

Let

$$L_{K,a} \sim D_1 \mid \mathbf{D}^T \mathbf{D} \leq a,$$

where  $\mathbf{D} = (D_1, \dots, D_K)$  follows a  $K$ -dimensional standard Normal distribution; let  $\varepsilon$  follow a univariate standard Normal distribution; and  $L_{K,a} \perp \varepsilon$ .

### Theorem 6.1

Under ReM with  $M \leq a$  and Condition 10, we have

$$\hat{\tau} - \tau \sim \sqrt{\text{var}(\hat{\tau})} \left\{ \sqrt{R^2} L_{K,a} + \sqrt{1 - R^2} \varepsilon \right\},$$

where

$$\text{var}(\hat{\tau}) = \frac{S^2(1)}{n_1} + \frac{S^2(0)}{n_0} - \frac{S^2(\tau)}{n}$$

is a Neyman (1923)'s variance formula and  $R^2 = \text{corr}^2(\hat{\tau}, \hat{\tau}_X)$  is the squared multiple correlation coefficient between  $\hat{\tau}$  and  $\hat{\tau}_X$ .

## 6.1 Rerandomization

Geometric interpretation of Thm. 6.1: TBA.

## 6.1 Rerandomization

### Proposition 6.1

Under the CRE, we have

$$R^2 = \text{corr}^2(\hat{\tau}, \hat{\tau}_X) = \frac{\frac{S^2(1 | X)}{n_1} + \frac{S^2(0 | X)}{n_0} - \frac{S^2(\tau | X)}{n}}{\frac{S^2(1)}{n_1} + \frac{S^2(0)}{n_0} - \frac{S^2(\tau)}{n}},$$

where  $\{S^2(1), S^2(0), S^2(\tau)\}$  are the finite population variances of  $\{Y_i(1), Y_i(0), \tau_i\}_{i=1}^n$ , and  $\{S^2(1 | x), S^2(0 | x), S^2(\tau | x)\}$  are the corresponding finite population variances of their linear projections on  $(1, X_i)$ .

## 6.1 Rerandomization

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# Proof of Eq. (1)

Neyman (1923) derives

$$\text{var}(\hat{\tau}) = \frac{S^2(1)}{n_1} + \frac{S^2(0)}{n_0} - \frac{S^2(\tau)}{n}.$$

In this case, we replace  $\hat{\tau}$  with  $\hat{\tau}_X$  and because  $X_i(0) = X_i(1) = X_i$ , then  $S_X^2(1) = S_X^2(0) = S_X^2$  and  $S_X^2(\tau) = S_X^2\{X(1) - X(0)\} = 0$ . Hence,

$$\text{cov}(\hat{\tau}_X) = \frac{1}{n_1} S_X^2 + \frac{1}{n_0} S_X^2. \quad \square$$

# Proof of Lemma 7

◀ Back

Neyman, J. (1923). On the application of probability theory to agricultural experiments. essay on principles. *Ann. Agricultural Sciences*, 1–51.